Victoria Tjhin

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Education

University of Waterloo - BASc, Faculty of Engineering, Honors in Nanotechnology Engineering | 2024 - 2029 (exp.)
- GPA: 3.92/4.00, awarded Consecutive Term Distinction
- Chemical Engineering (2024 - 2025)

Technical Skills

Languages Python, C++, TypeScript, SQL, MATLAB, Verilog, SystemVerilog
Frameworks & Libraries React Native, Node.js, Hono, NumPy, SciPy, Pandas, Matplotlib, YOLO, Intel Pin
Tools Git, Linux, Make, PostgreSQL, Twilio, GTKWave, ModelSim/VCS, Vivado, KiCad

Work Experience

Institute for Quantum Computing (IQC) - Research Assistant Jan 2025 - Present | Waterloo, ON
- Built a Python pipeline using NumPy and SciPy to automate UV-Vis analysis, reducing processing time by 90%.
- Automated data validation, visualization, and reporting workflow, reducing research workload up to 6 hours per week.

Ontario Battery and Electrochemical Research Centre (OBEC) - Battery Intern May - Aug 2026 | Waterloo, ON
- Automated Python workflows for battery cycling and impedance datasets, standardizing analysis across five cell builds.

Scotiabank - Workplace Transformation Analyst Jan - Apr 2025 | Toronto, ON
- Queried operational datasets using SQL and advanced Excel function across 6 ecosystems, identifying inefficiencies and supporting company's sustainment strategy.
- Analyzed utilization data for 13,000+ employees, identifying 34 underutilized workspaces and delivering data-driven recommendations that secured executive approval for strategic retrofit.

Projects

SafeStep - AI-Powered Accessibility & Safety App | React Native, TypeScript, Python, Node.js, YOLO, Twilio
- Built a cross-platform accessibility app for blind and low-vision users, integrating YOLO-based object detection to support environmental awareness and safer navigation.
- Developed an automated SMS alert workflow to notify designated caregivers when patients fail to respond to a voice-based fall check-in within 10 seconds.
- Optimized accelerometer based fall detection algorithms, reducing false positives by 30% through motion threshold tuning and improved false alarm handling.

Cycle-Accurate Cache Simulator | C++, Intel Pin
- Developed a cycle-accurate C++ cache simulator supporting configurable associativity, eviction, and write-allocation policies.
- Analyzed binary execution using Intel Pin to measure cache hit rates, AMAT, and cycle counts and optimized address decoding and memory access, reducing runtime 4x.

EMG Wearable Sensing System - Biotron | Embedded C++, ESP32
- Developed C++ firmware for ESP32-based sensor sustaining 1 kHz sensor sampling over 8-hour runs without resets.
- Debugged timing and data-integrity issues and optimized sensor data acquisition using Python, reducing dropped samples from 12% to 1% and improving signal-to-noise ratio by 3x.

ASIC Optical Autoalignment System - SSCS 2026 Chipathon | RTL, DSP
- Architected modular software block interfaces for the control system, enabling 3 sub-teams build and test components independently.
- Implemented parallel X/Y DSP pipeline to estimate peak offset in 2D and update position every dither cycle at 50 MHz.
- Restructured intermodule handshaking between sensing, DSP, and actuator blocks, reducing end-to-end latency by 40%.

Conductive Hydrodynamic Ink Printer - UWNRG | C++, Marlin firmware
- Developed C++ motion control firmware to synchronize three axes and lead screw logic within 50 µm of target.
- Debugged and validated firmware-to-motor communication during hardware bring up, coordinating motion across the system.
- Integrated motion control firmware with custom stepper driver hardware, reducing motion system backlash by 30% through control system and mechanical optimization.

Ethernet MAC Receiver Pipeline - UWASIC | SystemVerilog, Verification
- Developed SystemVerilog verification testbenches through randomized and directed frame, identifying CRC defects before integration.
- Pipelined CRC-32 processing engine with keep-aware partial-beat handling, carrying 64-bit beats at 156.25 MHz for a 10 Gb/s rate.

SPI-Controlled PWM Peripheral - UWASIC | Verilog, Tiny Tapeout Submission
- Implemented a register mapped SPI slave and 8-bit PWM generator in Verilog with duty and output-enable control.
- Verified timing and data integrity by building validation tests across 256 duty cycle settings across a 100 kHz SPI bus and 10 MHz clock.